

1. The Schur test (*presentation = 3 points, as for a section*)

Let $1 < p < \infty$, and set $q = \frac{p}{p-1}$, i.e. $\frac{1}{p} + \frac{1}{q} = 1$. Let $K : \mathbb{R}^2 \rightarrow \mathbb{C}$ be measurable w.r.t. 2-dimensional Lebesgue measure (*if you are not familiar with that, assume K is continuous*). Assume there exist $C > 0$ and $h : \mathbb{R} \rightarrow \mathbb{C}$ measurable with $h > 0$ a.e. (*h continuous with finitely many zeros*) such that

$$\int_{-\infty}^{\infty} |K(x, y)| h^q(y) dy \leq C h^q(x), \quad \text{a.e. (or for all } x),$$

and

$$\int_{-\infty}^{\infty} |K(x, y)| h^p(x) dx \leq C h^p(y), \quad \text{a.e. (or for all } x).$$

Prove that the linear map $T_K : L^p(\mathbb{R}) \rightarrow L^p(\mathbb{R})$, with

$$T_K f(x) = \int_{-\infty}^{\infty} K(x, y) f(y) dy, \quad f \in L^p(\mathbb{R}),$$

is well defined and bounded with $\|T_K\| \leq C$.

Hint: Apply Hölder to obtain

$$\begin{aligned} |T_K f(x)|^p &\leq \left(\int_{-\infty}^{\infty} |K(x, y)| h^q(y) dy \right)^{p/q} \left(\int_{-\infty}^{\infty} |K(x, y)| |f h^{-1}|^p(y) dy \right) \\ &\leq C^{p/q} h^p(x) \left(\int_{-\infty}^{\infty} |K(x, y)| |f h^{-1}|^p(y) dy \right). \end{aligned}$$

Integrate in x , use Fubini and the second assumption.

Remark The above proof shows that the result holds for any L^p -space, in particular for l^p .

2. Use the Schur test to prove that for $1 < p < \infty$:

a) The linear map $H : L^p(\mathbb{R}) \rightarrow L^p((0, \infty))$, with

$$Hf(x) = \frac{1}{x} \int_0^x f(t) dt, \quad f \in L^p((0, \infty)),$$

is well defined and bounded (*Hardy operator*).

b) The linear map $C : l^p \rightarrow l^p$ with

$$C(x_n) = \left(\frac{1}{n} \sum_{k=1}^n x_k \right), \quad (x_n) \in l^p,$$

is well defined and bounded (*Cesáro operator*)

3. Use the Schur test to prove that the Laplace transform defined in 16.3.3 is bounded on $L^2((0, \infty))$ (do not bother with the value of its norm).

4. Solve exercise 3 on page 184.

5. Use the Schur test to prove that the Hilbert-Hankel operator defined on Theorem 11, page 185 is bounded on $L^p((0, \infty))$, $1 < p < \infty$ (do not bother with the value of its norm).